CHANG-TENG LIN

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Alpha Researcher @ Kronos Research | Crypto HFT | Quantitative Research | C++ | Agentic AI

M.S. in Physics, National Taiwan University

B.S. in Physics, Minor in Mathematics, National Taiwan University

WORK EXPERIENCE

- **Kronos Research** July 2025 – Present

  ALPHA RESEARCHER, CRYPTO HIGH-FREQUENCY TRADING (HFT) / SYSTEMATIC TRADING

  - ✓ Conduct quantitative research on cryptocurrency market microstructure using large-scale, high-frequency tick-level market data; apply time-series and statistical analysis, feature engineering, and signal research to identify short-horizon predictive signals.
  - ✓ Independently conceived two HFT alpha signals from scratch and took them through hypothesis generation, feature/signal development, backtesting, C++ implementation, production deployment, and live trading.
  - ✓ Own the end-to-end research-to-production lifecycle, including experiment design, data analysis, alpha generation, signal validation, robustness evaluation, and production-quality strategy implementation; use Claude Code and agentic research workflows to accelerate research iteration.
  - ✓ Build backtests, simulations, and diagnostics to evaluate signal strength, stability, turnover, implementation sensitivity, and real-world performance before deployment.
  - ✓ Use strategy diagnostics and performance attribution to analyze production behavior, identify sources of performance degradation, and refine signal logic and execution decisions.
  - ✓ Improved existing systematic HFT strategies through iterative strategy optimization and empirical research, increasing Sharpe ratio and PnL.
  - ✓ Research both maker and taker execution at tick resolution, evaluating market-microstructure and order-flow effects including liquidity, transaction costs, slippage, and adverse selection.
  - ✓ Develop and optimize high-performance C++ for tick-data processing, alpha research, and trading-strategy implementation; used profiling and performance optimization to achieve a 50% speedup in alpha calculation.

RESEARCH EXPERIENCE

- **Tensor Network** *Prof. Ying-Jer Kao*

  GRADUATE RESEARCH, DEPT. OF PHYSICS, NATIONAL TAIWAN UNIVERSITY Summer 2022 – 2024

  - ✓ Thesis: Generating Function Study of Kitaev Model Spectral Function
  - ✓ Developed and extended tensor-network methods for two-dimensional ground-state and excitation calculations.
  - ✓ Implemented numerical algorithms for two-dimensional ground-state and excitation problems and extended them to more complex systems.
  - ✓ Maintained Cytnx, a tensor-network library for physics simulation; wrote C++/Python code, algorithms, data structures, and unit tests.
  - ✓ Profiled and optimized performance-critical numerical simulation code, improving computational efficiency for large-scale research workloads.

- **Spintronics** *Prof. Ssu-Yen Huang*

  UNDERGRADUATE RESEARCH, DEPT. OF PHYSICS, NATIONAL TAIWAN UNIVERSITY Fall 2021 – Spring 2022

  - ✓ Studied spin-orbit torque and spin Hall angle using NM/FM multilayers, harmonic measurements, and lock-in amplification.

- ✓ Fabricated nanometer thin films and performed PPMS magnetoresistance/Hall measurements and rapid thermal annealing (RTA).

TECHNICAL SKILLS

- **Programming & Systems** 2013 – Present
  - ✓ C++ (primary), Python, CUDA, Rust, Julia, Java, and JavaScript; production-quality C++ for quantitative research, trading strategies, and performance-critical systems.
  - ✓ Linux, Git, VTune, profiling/debugging tools, shell scripting, high-performance computing (HPC), performance optimization, and systems-level problem solving.
- **Quantitative Research & Systematic Trading** 2025 – Present
  - ✓ High-frequency trading (HFT), systematic trading, quantitative research, alpha research/generation, signal generation, predictive signals, feature research/engineering, market microstructure, and intraday/short-horizon strategy development.
  - ✓ Backtesting, simulation, experiment design, statistical analysis/modeling, time-series analysis, signal validation, robustness testing, performance attribution, strategy optimization, and research-to-production workflows.
- **Market Data, Execution & Production Trading** 2025 – Present
  - ✓ High-frequency market data, tick-level data, real-time data feeds, large datasets, market-data analysis, order flow, maker/taker strategies, execution analysis, and live/production trading.
  - ✓ Execution-quality concepts including liquidity, transaction costs, slippage, adverse selection, implementation sensitivity, PnL, Sharpe ratio, and risk/performance evaluation.
- **Data Science, Statistics & Machine Learning** 2022 – Present
  - ✓ NumPy, SciPy, Pandas, Matplotlib, PyTorch, and Keras for data processing, visualization, statistical inference, model building, feature engineering, machine learning, and experiment evaluation.
  - ✓ Experience analyzing noisy/high-dimensional datasets and building reproducible numerical research workflows in both trading and graduate research.
- **Algorithms & Mathematical Foundations** 2018 – Present
  - ✓ Probability theory, statistics, mathematical finance, linear algebra, analysis, ODE/PDE, numerical methods, and mathematical modeling for quantitative research and data analysis.
  - ✓ Competitive programming and algorithmic problem solving: dynamic programming, binary search, Fenwick/segment trees, graphs, number theory, and divide and conquer.
- **Scientific Computing & Performance Engineering** 2022 – Present
  - ✓ C++/CUDA numerical computing, profiling, performance engineering, high-performance computing, and large-scale simulation; experience optimizing performance-critical research code.
  - ✓ Tensor-network computation and high-dimensional numerical simulation; maintained the Cytnx C++/Python library with algorithms, data structures, and unit tests.

AGENTIC AI & RESEARCH ENGINEERING

- **Claude Code & Agentic Workflows** 2025 – Present
  - ✓ Extensively use Claude Code, AI coding agents, agentic coding, agentic software engineering, LLM agents, agentic research workflows, research automation, agent orchestration, tool orchestration, and human-in-the-loop workflows.
  - ✓ Apply context engineering, prompt engineering, harness engineering, and loop engineering to structure long-horizon agent workflows, sub-agent/multi-agent research, task decomposition, evaluation/verification loops, and iterative execution.
  - ✓ Design and use reusable agentic research frameworks for hypothesis exploration, code generation/refactoring, testing, debugging, data analysis, experiment execution, validation, and research-to-production iteration.

PUBLICATION

Kai-Hsin Wu, **Chang-Teng Lin**, Ke Hsu, Hao-Ti Hung, Manuel Schneider, Chia-Min Chung, Ying-Jer Kao, Pochung Chen: “The Cytnx Library for Tensor Networks,” 2024.